MIL-G-52880A
7 October 1983
SUPERSEDING
MIL-G-52880(ME)
7 August 1975

MILITARY SPECIFICATION

GENERATOR SET, DIESEL ENGINE, 500 KW

50/60 HERTZ, TACTICAL UTILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a skid mounted, unhoused, tactical utility generator (MIL-STD-1332) set rated for 500 kilowatts (kW), 60 Hertz (Hz), 0.8 power factor (pf) lagging, at 1800 revolutions per minute (rpm), or 417 kW, 50 Hz, 0.8 pf lagging, at 1500 rpm. The generator set shall be reconnectable for the following nominal voltage ratings:

a. 3 phase, 4 wire, 240 volts line-to-neutral and 416 volts line-to-line (240/416 volts).
b. 3 phase, 4 wire, 120 volts line-to-neutral and 208 volts line-to-line (120/208 volts).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6115

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SPECIFICATIONS

FEDERAL

VV-F-800	- Fuel Oil, Diesel.
ZZ-B-190	- Belts, V: Engine Accessory Drive; Minus 40° F, General Specification For.

MILITARY

MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-L-9000	- Lubricating Oil, Shipboard Internal Combustion Engine, High Output Diesel.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-A-11755	- Antifreeze, Arctic-Type.
MIL-F-16884	- Fuel Oil, Diesel, Marine.
MIL-T-22085	- Tape, Adhesive, Preservation and Sealing.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-G-28554	- Generator Sets, Mobile Electric Power and Supplemental Equipment, Packaging of.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engines, Arctic.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-195	- Marking of Connections for Electric Assemblies.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462	- Electromagnetic Inteference Characteristics Measurement of.
MIL-STD-705	- Generator Sets, Engine-Driven, Methods of Tests and Instructions.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-1332	- Definitions of Tactical, Prime, Precise, and Utility Terminologies for Classification of The DOD Mobile Electric Power Engine Generator Set Family.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

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DRAWINGS

USAF/SMAMA

TA 30554.76-029	- Generator Set - Diesel Eng. 500KW, 50/60 Hz, Tactical, Utility.
TA 30554.76-11000	- Housing Kit.
TA 30554.76-11339	- Remote Control Module.
Dwg. 81-11022	- Automatic Control Module.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. This specification covers a skid mounted, tactical utility generator set rated for 500 kilowatts (kW), 60 Hertz (Hz), 0.8 power factor (pf) lagging, at 1800 revolutions per minute (rpm), or 417 kW, 50 Hz, 0.8 pf lagging, at 1500 rpm, complete with accessories and auxiliary equipment as specified herein and required for normal operation. The generator set shall be rated for continuous duty operation and shall be capable of a continuous 2 hour overload at 110 percent of the 60 Hz rating from -25° F (-31.7° C) to +125° F (51.7° C) and at altitudes from sea level to 1500 feet (719.8 mmHg). Generator and auxiliaries shall be rated for 120/208 and 240/416 volts, 0.8 pf lagging, 50/60 Hz. When set is in automatic mode, the set shall be capable of starting, stopping, monitoring operation, automatic synchronizing and parallel operation with other sets of like characteristics and complete control from the set control panel or from the remote control module. When set is in manual mode, the set shall be capable of starting, stopping, monitoring operation, manual synchronizing and parallel operation with a utility and other sets of like characteristics and complete control from the remote control module. When specified (see 6.2), the generator set with automatic control system shall provide for unattended operation. The set shall be fabricated in accordance with top assembly TA 30554.76-029.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed drawing requirements is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the set. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.1.2 Reliability. The specified Mean-Time-Between-Failure (MTBF) shall be 500 hours when tested as specified in 4.8.1.

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3.2 First article. The contractor shall furnish two complete assembled sets for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his company can produce sets that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.4).

3.3 Drivebelts. Drivebelts shall conform to ZZ-B-190.

3.4 Environmental, operational, and handling. When equipped with all accessories and kits, the set shall start (see 6.3.1), and operate as follows.

3.4.1 Starting.

3.4.1.1 With engine preheat system. After the engine preheat system has been in operation for not more than six (6) hours, the set shall start:

a. At ambient temperatures from +125° Fahrenheit (F) [51.7° Centigrade (C)] to +32° F (0.0° C) at mean sea level (msl) and at any possible relative humidity.
b. With 12 inches of rain per hour impinging on the set at angles from the vertical up to 45 degrees from vertical when housing kit (see 3.7) is installed.
c. With the base of the set in planes from level to 5 degrees from level.
d. At any altitude (see 6.3.16) from sea level [760 millimeters of mercury (mmHg)] and a maximum temperature of 125° F (51.7° C) up to and including 8,000 feet (564.4 mmHg) and a maximum temperature of 95° F (35.0° C).

3.4.1.2 With the housing kit installed. The set shall meet the requirements of 3.4.1.1 at ambient temperatures from 125° F (51.7° C) to -25° F (-31.7° C) when the housing kit is installed (see 3.7).

3.4.2 Storage temperature and humidity. The set shall not be damaged (see 6.3.3) by storage in temperatures from +155° F (68.3° C) to -65° F (-53.9° C) at any possible relative humidity within that range and shall not be damaged by exposure to five consecutive 48 hour cycles as specified in MIl-STD-705, method 711.1.

3.4.3 Operating. After the set has started, the set shall operate for a total of 5,000 hours in accordance with the provisions specified herein and under conditions as follows:

a. Without maintenance, overhaul, or replacement of parts, other than lubrication, routine engine servicing, and periodic adjustments as permitted in the schedule of table I.
b. At continuous loads up to and including rated load under the conditions specified in 3.4.1.1, except for the altitude conditions (see 3.4.1.1(d)).

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c. For the altitude condition, the set shall meet the above at rated load and at 110 percent rated load for 2 hours at an altitude of 1,500 feet (719.8 mmHg) at +125° F (+51.7° C), not less than 80 percent of rated load at 5,000 feet (632.4 mmHg) at +107° F (+41.7° C), and not less than 75 percent of rated load at 8,000 feet (564.4 mmHg) at +95° F (+35° C).

3.4.4 Operating speeds. The operating (rated) speed of the generator set shall be 1800 rpm for 60 Hz operation and 1500 rpm for 50 Hz operation. The set shall operate at speeds up to and including 125 percent of 60 Hz rated speed for a period of not less than 5 minutes without damage (see 6.3.18).

3.4.5 Flexural vibrations and critical speeds. The set shall be free from dangerous flexural vibration (see 6.3.4), and dangerous torsional vibrations (see 6.3.5), between 80 and 125 percent of rated speed for both 50 Hz and 60 Hz operation (see 3.4.4).

3.4.6 Rough handling. The set shall withstand, without damage (see 6.3.6), rough handling which would normally be encountered during rail, or highway trailer transportation (see 6.3.7).

Table I. Servicing, adjustment, and replacement test schedule*.

No.	Minimum Hrs. of Operation	Servicing, Adjustment, and Replacement Allowed
1	8	Check and add lubricating oil, without stopping engine, as required.
2	8	Add coolant, without stopping engine, as required.
3	100	Check air filter restriction indicator. Change or clean filter panels, when required.
4	250	Change lubricating oil filters and lubricate accessories as recommended by the engine manufacturer.
5	500	Inspect engine pistons, rings and cylinder liners to the extent possible without disassembly.
6	500	Adjust belt tension.
7	1,000	Change fuel oil filters. Drain fuel tank condensation.
8	250	Perform oil analysis and change lubricating oil if required.
9	500	Check battery, battery charging system, all terminals and connections. Add water to cells as required.
10	2,500	Check valve tappet clearance and reset, if necessary.
11	2,500	Clean injection nozzles, clean and adjust fuel setting as necessary.
12	2,500	Replace belts as necessary.
13	5,000	Inspect governor, change oil, lubricate and adjust governor linkage.
14	10,000	Replace fuel injectors, as necessary.
15	250	Replace water filter.

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*This schedule shall be followed during first article inspection and quality conformance tests. If maintenance shutdown, as permitted in the schedule, should become due (because of hours accrued during previous testing) after the beginning of the endurance reliability test, the shutdown shall be postponed until the end of a 100 hour cycle of endurance-reliability test.

3.4.7 Maximum power. The maximum power of the set in kilowatts shall be not less than 530 Kw when operating on diesel fuel (see 3.8.2) except grade JP-5.

3.5 Sound pressure levels. The set shall not exceed the audio noise limits specified in table IA:

TABLE IA. Maximum SPLs in Decibels (dB) (Reference 0.0002 Microbar).

	Octave Band Midfrequency (Hz)							
Receiver Location	63	125	250	500	1000	2000	4000	8000
External: (100 ft 62 radial distance)	84	82	79	77	72	72	7	

3.6 Air pollution control. The generator set shall comply with all Federal regulations covering the control of air pollution.

3.7 Housing kit. When specified (see 6.2), a housing kit shall be furnished and installed, which shall include a housing, air intake louvers, shutters, and grills in accordance with TA-76-11000.

3.8 Engine.

3.8.1 Fuels and lubricants. The set shall be capable of satisfactorily starting and operating on fuels conforming to VV-F-800, Grades DF-A, DF-1 and DF-2, MIL-F-16884, and turbine fuel conforming to MIL-T-5624, Grade JP-5. The set shall be capable of continuous operation using lubricating oil conforming to MIL-L-9000 and MIL-L-2104 and MIL-L-46167.

3.8.2 Coolant. Cooling compound shall conform to MIL-A-11755.

3.9 Governor system.

3.9.1 Governor system performance. When initially adjusted for zero droop, the governing system shall provide set performance, as specified herein, without adjustments other than permitted (see 3.4.3)

3.9.1.1 Frequency regulation. The frequency regulation see (6.3.10) for zero droop shall not exceed one-fourth of 1 percent of rated frequency for every load change from no load to rated load.

3.9.1.2 Short-term frequency stability (30 seconds). At every constant load from no load to rated load, the speed governing system shall maintain frequency

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within a bandwidth equal to 0.5 percent of rated frequency. The governing system shall not permit repetitive periodic frequency variation, commonly called hunting, even though within the allowable 0.5 percent band.

3.9.1.3 Long-term frequency stability (4 hours). At constant ambient temperature, constant barometric pressure, constant voltage, and any constant load from no load to rated load, the speed governing system shall maintain frequency within a bandwidth equal to 1.0 percent of rated frequency within a 4-hour operating period.

3.9.1.4 Frequency drift. With the set operating at constant load and voltage, a change in the ambient temperature of 60F° (33.3C°) in an 8-hour period shall not cause the frequency to change by more than 1 percent, set temperature stabilization being accomplished at both the initial and final ambient temperature conditions.

3.9.1.5 Frequency transient performance. With the speed control system set for 0 to 3 percent operation, and following a sudden 50 percent change in load between no load rated load and vise-versa, the speed governing system shall reestablish stable engine operating conditions (See 6.3.11) within 4 seconds. The maximum transient frequency change above the new steady state frequency (overshoot) shall be not more than 4 percent of rated frequency. The maximum transient frequency change below the new steady state frequency (undershoot) shall be not more than 4 percent of rated frequency (see 6.3.12).

3.9.1.6 Frequency adjustment. It shall be possible to adjust the frequency when the set is operating as follows:

a. At any load up to and including rated load:

(1) Between 57.5 and 62.5 but not above 65 Hz for the 60 Hz operating condition.
(2) Between 47.5 and 52.5 but not below 45 Hz for the 50 Hz operating condition.

A selector switch shall be used to transfer from 50 to 60 Hz operation. The frequency adjusting control shall be such that it is not possible to adjust the frequency for operation within the 47.5 and 52.5 Hz range when the switch is in the 60 Hz position; and it shall not be possible to adjust for operation in the 57.5 and 62.5 Hz range when the switch is in the 50 Hz position.

3.10 Saturation curves. The curves listed herein shall be plotted with the generator armature terminal volts on the vertical axis versus the exciter field current on the horizontal axis:

a. Open circuit saturation curve.
b. Synchronous impedence curve (short circuit saturation curve).
c. Zero power factor saturation curve.
d. Rated load current saturation curve.

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The exciter field current shall be:

1 ampere at no load 60 Hz
2.7 amperes at full load 60 Hz
1.5 amperes at no load 50 Hz
3.2 amperes at full load 50 Hz

3.11 Generator. The generator shall withstand operation at 2250 rpm for 15 minutes without damage (see 6.3.18).

3.11.1 Winding resistance. The resistance of each winding shall be as specified on the applicable drawings. All resistance values shall be corrected to a temperature of 77° F (25° C).

3.12 Generator performance. The generator together with the excitation system shall perform as follows:

3.12.1 Insulation resistance. Insulation resistance shall not be less than 1 megohn in an ambient temperature of 77° F (25° C).

3.12.2. Temperature rise. The temperature rise shall not exceed 100° C (171° F), for class F insulation, when measured by the resistance method.

3.12.3 Short Circuit. The generator, exciter, and voltage regulator operating as a unit shall withstand the following short circuits at its output terminals, when operating at rated load, without reduction of the dielectric strength to a point where it will not meet the requirements of 3.12.7.

a. A 10 second symetrical three phase short circuit with a sustained short circuit current of not less than 300 percent of rated output current.
b. Line to line short circuits of sufficient duration to produce an I^2T value of not less than 160, using the current at time T. The current at time T shall not be less than 300 percent of rated output current.
c. Line to neutral short circuits of sufficient duration to produce an I^2T value of not less than 200, using the current at time T. The current at time T shall not be less than 300 percent of rated output current.

3.12.4 Transient reactance, rated voltage. The direct axis transient reactance shall be not more than 25 percent for the 60 Hz operation.

3.12.5 Voltage waveform. The deviation factor for the line-to-line and line-to-neutral voltage for each voltage connection shall be not more than 5 percent, nor shall any single voltage harmonic be more than 2 percent under the following output conditions of the generator, when under control of its exciter and voltage regulator:

a. No load.
b. Up to and including rated kW at any power factor between unity and 0.80 lagging.

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In addition, when the waveform is viewed on an oscilloscope having a bandwidth of DC - 1.5 mHz and a useable viewing screen of 8 x 10 CM, there shall be no evident discontinuities, spikes, or notches in the waveform when oscilloscope gain is set such that peak-to-peak excursion of the voltage wave covers approximately the full vertical dimension of the viewing screen.

3.12.6 Voltage unbalance. Maximum difference between line-to-line voltages shall be not more than 5 percent of rated voltage for the 240/416 volt connection under the condition of a single-phase, line-to-line, unity-pf load of 25 percent of rated current and no other load on the generator. This requirement shall apply with the generator under control of its exciter and voltage regulator.

3.12.7 Dielectric strength. Electric windings shall withstand the following 60 Hz voltages applied for 1 minute:

a. Generator armature - 1,832 volts between phase windings and 1480 volts between phase windings and ground.
b. Generator field and exciter windings - 10 times ceiling voltage, but neither less than 1,500 nor more than 3,500 volts. (Applied between windings and ground.)
c. Windings energized by the 28 volt dc control, cranking and battery-charging systems -500 volts. (Applied between windings and ground.)
d. All others - Twice rated voltage plus 1000 volts. (Applied between windings and ground and between windings where applicable.)

3.12.8 Phase balance (voltage). With the generator under control of its exciter and voltage regulator, and operating at rated voltage, frequency, and no load, the maximum difference in the three line-to-neutral voltages shall be not more than 1 percent of rated line-to-neutral voltage. The maximum difference between the voltages of the 120-volt windings of any one phase shall be not more than 1 volt.

3.12.9 Voltage regulation. The voltage regulation (see 6.3.10) from no load to rated load and from rated load to no load shall be not more than 2 percent of the rated voltage.

3.12.10 Short term stability (30 seconds). At any constant load from no load to rated load, the voltage at the set terminals shall remain within a bandwidth equal to 1 percent of rated voltage.

3.12.11 Long term stability (4 hours). At constant ambient temperature, constant barometric pressure, constant frequency and any constant load from no load to rated load, the voltage shall remain within a bandwidtn of 2 percent of rated voltage in a 4-hour operating period.

3.12.12 Drift. With the set operating at constant load and frequency, a change in ambient temperature up to 60 F° (33.3 C°) in an 8-hour period shall not cause the voltage to change by more than 1 percent, set temperature stabilization being accomplished at both the initial and final ambient temperature conditions.

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3.12.13 Voltage-transient performance. With the reactive load compensation of the voltage regulator inoperative, and with the engine speed control system set for 0 to 3 percent regulation, performance of the set under transient conditions as measured by an oscillograph shall be as follows:

a. With the set initially operating at rated frequency and rated voltage, and following any sudden 50 percent increase in load between no load and rated load, the instantaneous root mean square voltage shall not drop to less than 80 percent of rated voltage and shall reach stable conditions within 3 seconds.
b. With the set initially operating at rated frequency, rated voltage and rated load, and following any sudden change in load from rated load to no load, the initial voltage transient shall involve a voltage rise not to exceed 130 percent of rated voltage and shall reach stable conditions within 3 seconds.

3.12.14 Voltage operating range. The set shall operate at any loads between no load and rated load at the following line-to-line voltages:

a. Between 395 and 500 volts for the 240/416 volt connection and between 197 and 250 volts for the 120/208 volt connection for 60 Hz operation.
b. Between 380 and 426 volts for the 240/415 volt connection and between 190 and 213 volts for the 120/208 volt connection for 50 Hz operation.

3.13 Parallel operation. The set shall be capable of parallel operation at zero speed droop with not more than 2 percent voltage droop, and at 3 percent speed droop with not more than 3 percent voltage droop. The governor parallel control and reactive load compensation control shall remain in the same position during both conditions.

3.13.1 Parallel operation with zero speed droop. With their governor and voltage regulator paralleling circuits properly interconnected, any four like sets operated in parallel shall divide load in accordance with the following, as system load at rated power factor is varied between 0 and 100 percent (and vice versa) of the combined rating of the connected sets:

a. Real power division. The difference between the average individual kW output shall at no time be more than 10 percent of the kW rating of one set.
b. Power exchange. At any constant system load, power exchange between the sets shall not exceed 10 percent of the rating of one set. (Power exchange is the difference between the maximum and minimum power output of one set, for constant system load conditions, as determined from oscillographic measurements.)
c. The reactive power division. The difference between the average reactive kilovolt-amperes (kvar) supplied shall at no time be greater than 10 percent of the reactive kvar rating of one set.

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For the above requirement, the initial system load, both kW and kvar, shall be equally divided between the sets. Thereafter, no adjustment shall be made to the governors, voltage regulators, or other components of the set as the system load is varied.

3.13.2 Parallel operation with speed droop. With the governor droop switch in the droop position and the voltage regulator shorting plug installed in the paralleling receptacle, any four like sets operated in parallel shall divide load in accordance with 3.13.1, as system load at rated power factor is varied between 0 and 100 percent (and vice versa) of the combined rating of the connected sets.

3.14 Output bus. Buses shall be conspicuously marked and identified in accordance with MIL-STD-195. The phase sequence shall be A, B, C.

3.15 Indicating insturments. Instruments shall be in accordance with applicable drawings.

3.16 Annunciator alarm system. The annunciator alarm system shall consist of audible, not less than 10 dB above operating ambient, and visual signals to attract the attention of the operator and indicate the occurrence of abnormal conditions in the set. The visual alarm shall consist of a backlighted annunciator panel with a window for each function specified herein. The annunciator alarm system shall have three pushbuttons or momentary switches and marked: "ALARM SILENCE", "TEST", AND "RESET", respectively. When a monitoring or protective device operates, the annunciator alarm system shall indicate the condition by sounding the audible alarm and by a flashing light behind the respective window(s) of the annunciator panel. Operation of the ALARM SILENCE pushbutton shall silence the audible alarm and cause the light(s) to stop flashing, but remain lighted. After the cause of alarm has been corrected, operation of the RESET pushbutton shall extinguish the light(s) and reset the system. Operation of the TEST pushbutton shall sound the audible alarm and light all windows of the annunciator panel as long as the pushbutton is held in the depressed position. The operations to be indicated on the annunciator alarm system shall include the following:

a. "HIGH COOLANT TEMPERATURE" (see 3.17.1.2).
b. "HIGH LUBRICATING OIL TEMPERATURE" (see 3.17.1.5).
c. "LOW LUBRICATING OIL PRESSURE" (see 3.17.1.3).
d. "LOW FUEL LEVEL" (see 3.17.1.4).
e. "OVERSPEED" trip (see 3.17.1.1).
f. "OVERVOLTAGE" trip (see 3.17.2.1).
g. "UNDERVOLTAGE" trip (see 3.17.2.2).
h. "OVERLOAD" trip (see 3.17.2.4).
i. "REVERSE POWER" trip (see 3.17.2.6).
j. "SHORT CIRCUIT" trip (see 3.17.2.3).

3.17 Protection system. The set shall be equipped with protective devices to accomplish functions as described herein.

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3.17.1 Engine protective devices. The following devices shall act to simultaneously open the set's output circuit interrupter and shut down the engine.

3.17.1.1 Overspeed. The overspeed device shall act to open circuit interrupter and stop engine at 125 percent ± 5 percent of 60 Hz speed.

3.17.1.2 High coolant temperature. This device shall actuate a warning signal when the coolant temperature reaches 213° F ± 3° F and a shutdown when the temperature reaches 222° F ± 3° F.

3.17.1.3 Low oil pressure. This protective device shall actuate a warning signal when the oil pressure drops to 21 ± 2 psi and a shutdown when the oil pressure drops to 13 ± 2 psi.

3.17.1.4 Low fuel level. The low fuel level protective device shall actuate when the fuel in the tank reaches a level at which sufficient fuel remains for only 15 minutes operation of the set at rated load.

3.17.1.5 High oil temperature. This device shall actuate a warning when the engine oil reaches a temperature of 255° F ± 3° F and a shutdown when the temperature reaches 265° F ± 3° F.

3.17.2 Other protective devices. The following devices, associated with the alternator or electrical output parameters, shall be provided. These devices shall operate only to open the set output circuit interrupter, except where otherwise required. All protective devices except synchronizing check relay and undervoltage relay shall be manual reset.

3.17.2.1 Overvoltage. An overvoltage protection device shall actuate in not more than 1 second after the set output voltage has risen to and remained at any value greater than 153 ± 3 volts for not less than 200 milliseconds. The overvoltage device shall also simultaneously stop the engine and signal annunciator.

3.17.2.2 Undervoltage. This device shall operate instantaneously when the set output voltage drops below 48 volts. It shall not operate instantaneously at voltages above 72 volts. In addition, the device shall operate in 6 ± 2 seconds after the set output voltage has fallen to and remained at not more than 99 ± 4 volts and signal annunciator.

3.17.2.3 Short circuit. This protection device shall operate instantaneously in the event set output current in any phase exceeds 425 ± 25 percent of rated current for both series and parallel connection of generator windings. These current values apply for the 60 Hz rating.

3.17.2.4 Overload. This protection device shall not trip unless current in any phase exceeds 110 percent of rated value; it shall trip within 8 ± 2 minutes for 130 percent of rated current and signal annunciator. These current values apply for the 60 Hz rating.

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3.17.2.5 Synchronizing check relay. The synchronizing check relay shall sense the voltage, frequency, and phase sequence between the generator and the bus (load side of the circuit breaker). The syncronizing check relay shall be electrically interlocked with the closing circuit of the circuit interrupter. The synchronizing check relay shall verify that the generator is synchronized with the bus before the circuit breaker can be closed.

3.17.2.6 Reverse power. This device shall actuate if power flow into the set exceeds 15 percent for 3 seconds.

3.18 Automatic control module (ACM). When specified (see 6.2) for automatic operation, one ACM shall be required for up to four generator sets for unattended operation. The ACM shall be constructed in accordance with drawing 81-11022 of TA 30554.76-029.

3.19 Remote control module. When specified (see 6.2), each generator set shall be furnished with a separate remote control module (Ref TA30554.76-11339), and remote control cable (see 3.20). When used, the remote module and cable will allow remote start and stop, monitor, control of the set in unit operation, and paralleling with like sets, from a remote location up to 1000 feet away.

3.20 Remote control cable (harness). When specified (see 6.2), each generator set shall be furnished with a remote control cable. Reference TA30554.76-11339.

3.21 Electromagnetic interference. The electromagnetic interference emission and susceptibility characteristics of the set shall not exceed the UM04 limits of class C2 equipment of MIL-STD-461 even though the set kVA rating is greater than that specified in the standard.

3.22 Air transportation. The set shall be suitable for transport as a palletized load on a C-130, C-141 and C-5A aircraft. Certification by the United States Air Force is required.

3.23 Delivery lubrication service. Unless otherwise preserved for packaging (see 5.1), all components shall be lubricated. Grease lubrication, including lubrication of sealed bearings, shall be with military greases conforming to MIL-G-10924 or MIL-G-23827, as applicable. The sets shall be serviced with winter grades of military oils designated for use in the temperature range of 0° F (-17.8° C) to 32° F (0°C). Each component shall be tagged in a conspicuous place to indicate the temperature range and grades of oils and greases used.

3.24 Lifting provisions. The set shall be provided with lifting provisions, slings, and spreader bars in accordance with applicable drawings.

3.25 Tiedowns. The set shall be provided with tiedown attachment points in accordance with applicable drawings.

3.26 Information and instruction plates. First article sets shall be delivered with all specified plates installed. For production sets, information

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and instruction plates may be installed after individual quality conformance tests (see 4.6) have been completed and the final overspray of paint is applied (see 3.27).

3.27 Treatment and painting. The set shall be treated and painted in accordance with the applicable drawings (see 3.1.1), except as specified herein. The following shall not be painted: Terminal wiring connections, governor linkage, instruction diagrams, nameplates, rectifiers, relays, circuit breakers, switches, hose, drivebelts, and all other parts whose operation or function would be adversely affected by paint. Treatment and painting shall be optional for the following if located within the set housing and not visible during operation of the set: Corrosion-resisting metals; metals treated to be corrosion resistant; and items treated and painted to be corrosion resistant by the original manufacturers, such as MS starting motor, ether tank and circuit interrupter. After assembly and testing, the interior of the set shall be retouched to correct any paint damage and the entire exterior of the set, except items adversely affected, shall be oversprayed.

3.28 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Disassembly inspection. Failure of any inspection of the first article generator, generator with excitation system, or sets shall be cause for disassembly, in the presence of a Government representative, to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article sets. Reassembly with acceptable components or parts and reinspection shall be the responsibility of the contractor.

4.1.2 Inspection failure. Should the Government elect to perform or repeat any inspection or test in this specification, failure of a set to meet any requirement specified herein shall be cause for refusal by the Government to accept production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on any sets previously produced under the contract. Any deficiencies

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found as the result of such elective inspection or test will be considered prima facie evidence that all sets accepted prior to the completion of such inspection or test are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer. The provisions of this paragraph apply, notwithstanding any prior acceptance of first articles or first article test reports.

4.1.3 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications, standards and drawings.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.6).
c. Inspection of packaging (see 4.9).
d. Comparison testing.

4.3 First article inspection. First article set(s) shall be examined and tested as specified herein. The inspection reports and inspection records of the generator set shall describe:

a. Malfunction, evidence of damage, failure, or adjustment, other than adjustments permitted in this specification and the applicable test methods, which occurs during the examination and tests.
b. The cause of the malfunction, damage, or failure and reason for the adjustment.
c. The corrective action taken.

The report covering inspection of the first article set shall be forwarded to the contracting officer upon completion of inspection.

4.3.1 Examination. Each set shall be examined, without disassembly for conformance to the drawings (see 3.1). Noncompliance with the drawings shall be cause for rejection.

4.3.2 Tests. Each set shall be subjected to the applicable tests marked "X" under column 1 of table II. For generator only tests, the tests shall be conducted prior to assembly into the set. Failure of any test shall be cause for rejection.

4.4 Generator sets. The sets shall be subjected to all the tests marked "X" under column 2 of table II after the first article generators have passed all tests marked "X" listed under column 1 of table II and the first article generator sets have passed the examinations required by 4.3.1. The tests shall be conducted in any order except that the rail impact shall be performed first, and the endurance and reliability test shall be the last performed.

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TABLE II Test Schedule.

First Article		Quality Conformance	Test	Test Method (MIL-STD-705) or Test paragraph (See Note 7)	Requirement Paragraph
Generators	Generator Sets	Individual Generator Set			
1	2	3	4	5	6
X	-	X	Insulation-resistance	301.1 (see Note 1) (isolate diodes).	3.12.1
X	-	-	High Potential	302.1 (see Note 1) (isolate diodes).	3.12.7
X	-	X	Winding-resistance	401.1.	3.11.1
X	-	-	Open circuit saturation curve	410.1 (at both 50 Hz & 60 Hz).	3.10
X	-	-	Short circuit saturation curve	411.1 (At both 50 Hz and 60 Hz.	3.10
X	-	-	Zero power factor saturation curve	412.1 (at both 50 Hz & 60 Hz).	3.10
X	-	-	Rated load saturation curve	413.1 (at both 50 Hz & 60 Hz).	3.10
X	-	-	Direct axis transient reactance	425.1.	3.12.4
-	X	-	Railroad impact	740.5 (See Note 6)	3.4.6 6.3.7.1
-	X	X	Start and Stop	503.1.	
-	X	X	Frequency and voltage regulation, stability and transient response (Short Term) Frequency and voltage stability (Long term)	608.1 and 608.2 (At both 50 and 60 Hz) (Load steps shall be NL-1/2L, 1/4L-3/4L, and 1/2L-4/4L. Each load shall be applied and removed four (4) times.	3.4 3.9.1
X	-	-	Overspeed (Generator only)	505.3 (At 125 percent rated 60 Hz speed for 15 minutes).	3.11

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TABLE II Test schedule. (Continued)

First Article		Quality Conformance	Test	Test Method (MIL-STD-705) or Test paragraph (See Note 7)	Requirement Paragraph
Generators	Generator Sets	Individual Generator Set			
1	2	3	4	5	6
-	X	X	Overspeed (Set)	505.1 (At 125 percent rated 60 Hz speed for 5 minutes).	3.4.4
-	X	X	Overspeed protection devices	4.8.15.	3.17.1.1 3.4.4
-	X	X	Phase balance (Voltage)	508.1 (To be measured at load terminals).	3.12.8
-	X	X	Circuit interrupter (Short circuit)	512.1 (Use oscillograph for preproduction sets only) (For production sets, L_1 - Lo, L_2 - Lo and L_3 - Lo shorts only, oscillograph need not be used).	3.16(j) 3.17.2.3
-	X	X	Circuit interrupter (Overload trip)	512.2.	3.17.2.4 3.16(h)
-	X	X	Circuit interrupter (Undervoltage and overvoltage)	512.3.	3.16(f) 3.17.2.1 3.16(g) 3.17.2.2
-	X	X	Indicating instrument (Electrical)	513.1 (For all voltage and frequency connections).	3.15
-	X	X	Low oil pressure protective device	515.1 (See Note 3).	3.17.1.3
-	X	X	Reverse power protective device	516.2.	3.17.2.6
-	X	X	Over temperature protective device	515.2 (See Note 3).	3.17.1.2

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TABLE II Test schedule. (Continued)

First Article		Quality Conformance	Test	Test Method (MIL-STD-705) or Test paragraph (See Note 7)	Requirement Paragraph
Generators	Generator Sets	Individual Generator Set			
1	2	3	4	5	6
-	X	X	Low fuel protective device	515.5 (See Note 3).	3.17.1.4
-	X	X	High oil temperature protective device	4.8.2 (See Note 3).	3.17.1.5
-	X	X	Regulator range	511.1 (At both 50 and 60 Hz).	3.12.14
-	X	X	Phase sequence (rotation)	507.1.	3.14
X	-	-	Generator power input	415.1 or 415.0.	3.1
X	-	-	Short circuit (mechanical strength)	625.1 Maintain the short circuit for 10 seconds. Perform the test for all possible line to line, line to neutral and the 3 phase conditions. Use the generator and excitation system.	3.17.2.3
-	X	X	Frequency adjustment range	511.2 (At both 50 and 60 Hz).	3.9.1.6
-	X	-	Voltage unbalance with unbalanced load (line-to-line)	620.2.	3.12.6
-	X	-	Voltage waveform (Harmonic analysis)	601.4 (At each line-to-line and line-to-neutral connection for all voltage and frequency ratings and the following load conditions (see Note 2):	3.12.5

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TABLE II Test schedule. (Continued)

First Article		Quality Conformance	Test	Test Method (MIL-STD-705) or Test paragraph (See Note 7)	Requirement Paragraph
Generators	Generator Sets	Individual Generator Set			
1	2	3	4	5	6
				a. No load b. Rated load at 0.8 pf lagging c. Rated load at 1.0 pf.	
-	X	-	Voltage dip for rated load	619.2 Perform at 50% of rated load. NL to 1/2L, 1/4L to 3/4L, 1/2L to 4/4L at both zero and 3% speed droop.	3.12.13
-	X	-	High temperature	710.1 (Take temperature rise measurements on the stator windings under the following conditions at 125° F (51.7° C) 50 Hz, 380 volts, 240/416 50 Hz, 426 volts, 240/416 60 Hz, 395 volts, 240/416 60 Hz, 500 volts, 240/416 Also perform method 619.2 at this condition. 1/4L - 3/4 and 1/2L - 4/4L.	3.12.2

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TABLE II Test schedule. (Continued)

First Article		Quality Conformance	Test	Test Method (MIL-STD-705) or Test paragraph (See Note 7)	Requirement Paragraph
Generators	Generator Sets	Individual Generator Set			
1	2	3	4	5	6
-	X	X	Parallel operation	630.1 (At both 50 and 60 Hz operation, for zero and 3% speed droop. All production sets shall be paralleled with the first article set in accordance with 4.8.4).	3.13
-	X	-	High temperature	710.1 at 125° F (With housing kit installed). Take temperature rise measurements on the stator windings at the conditions that gave the highest rise in the unhoused configuration at 125° F.	3.12.2
-	X	-	Remote control and operation	See 4.8.6.	3.19
-	X	-	Endurance-reliability	4.8.1 with housing kit installed.	
-	X	X	Maximum power (For gasoline and diesel generator sets)	640.1 (First article sets) and 640.4 (Production sets).	3.4.7
-	X	X	Annunciator alarm system	See 4.8.5.	
-	X	-	Remote operation	4.8.8 At 500 ft. from set.	

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TABLE II Test schedule. (Continued)

First Article		Quality Conformance	Test	Test Method (MIL-STD-705) or Test paragraph (See Note 7)	Requirement Paragraph
Generators	Generator Sets	Individual Generator Set			
1	2	3	4	5	6
-	X	-	Overload	4.8.13 At 125° F (51.7° C), (See Note 5).	3.1
-	X	-	Railroad impact (For optional equipment)	740.5 (See Note 8).	3.4.6 6.3.7.1
-	X	-	High temperature (For optional equipment)	4.8.11 At 125° F (51.7° C).	3.19 3.18
-	X	-	Humidity test (For optional equipment)	4.8.14.	3.18 3.19
-	X	-	Automatic control system	4.8.7.	3.18

TABLE III Schedule of optional tests.

Test	Test Method (MIL-STD-705), Test Paragraph, or as Required Below	Requirement Paragraph
1	2	3
Torsiographing	504.2	3.4.5
Rain	711.3 (with housing kit installed)	3.4.1.
Electromagnetic interference	MIL-STD-462 (at all rated frequency and voltage connections at no load rated load 1.0 PF and rated load 0.8 PF)	3.21.
Sound level	661.2 (see Notes 4 & 5)	3.5
Inclined operation	660.1 (5° only)	3.4.1.
Voltage and frequency drift	4.8.10	3.9.1.3 3.12.12.

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Table III Schedule of optional tests. (Continued)

Test	Test Method (MIL-STD-705), Test Paragraph, or as Required Below	Requirement Paragraph
1	2	3
Starting and operating (moderate cold-battery start)	701.2 at -25° F (-31.7° C). Use preheat system for 6 hours with housing kit installed. The start shall be made using the set control module, which has been at -25° F for at least 24 hours. Start shall be initiated by simulating a power failure.	3.4.1
Starting and operating (moderate cold-battery start)	701.2 at +32° F (0.0° C). Use preheat system for 6 hours. Do not perform methods 510.1, 511.1 and 513.2.	3.4.1.
Lifting and towing	740.4. Lift at 4 times the set weight, tow at 5 times the set weight. (see Note 5) Towing to be performed with housing kit installed only	3.24
Tiedown	See 4.8.3	3.25
High temperature at 125° C (5.7° C)	710.1 (Take temperature rise measurement on rotor windings only). (Slip rings required.)	3.12.2
Low temperature (for optional equipment)	4.8.12 at -25° F (-31.7° C)	3.4.1.
Automatic control system (sets designated for specific installation)	4.8.9	3.18.
Altitude operation	720.1c or 4.8.16 at 8,000 ft., 95° F (35° C) (see Note 5)	3.4.1.1
Humidity	711.1 (with housing kit installed).	3.4.2
Voltage waveform (oscillographic)	601.1 (at each line-to-line and line-to-Neutral correction for all voltage and frequency ratings and the following load corrections (see Note 2). a. No load. b. Rated load at .8 power factor lagging. c. Rated load at 1.0 factor lagging.	3.12.5
Air transportability	4.8.17	3.22 6.3.7.4

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NOTES

1. Insulation resistance and high potential tests may be made after assembly of the engine to the generator but before final assembly of the complete set.
2. Also, view waveform with oscilloscope. For individual tests, only method 601.4 need be used.
3. Also record the value at which the annunciator alarm sounds.
4. Exterior readings for sound level tests are to be taken at not less than eight positions, 45 degrees apart, on a 100-foot radial distance from the set. The interior readings for sound level tests are to be taken at not less than three positions located 2 feet from the front of the control center and not less than four positions in other work areas that are representative of required operator positions when the set is operating. Readings shall be taken at rated load and at no load with the set operating at rated voltage and frequency. The microphone height shall be 5 feet 8 inches and the ambient background noise level shall be not less than 6 dB below the specified noise level criteria being tested. Measurement shall be made in dB on the C scale.
5. Perform this test on both an unhoused set and a set with the housing kit installed.
6. The set shall be humped at speeds up to and including 8 mph. The set shall be wet with all liquids at operating level. The set shall be secured to the railroad car using the tiedown provisions. One set shall be tested with the engine end forward and the other set with the generator end forward. All reference to generator in test method 740.5 shall be changed to generator set.
7. Unless otherwise specified, all tests shall be conducted with the set in the 240/416 volt, 60 Hz configuration.

4.5 Inspection failure. Failure of the first article set to meet any requirement specified herein during and as a result of the examination and tests shall be cause for rejection of the first article set.

4.6 Quality conformance inspection.

4.6.1 Individual set inspection.

4.6.1.1 Examination. Each set shall be examined, without disassembly for conformance to the drawings (see 3.1). Noncompliance with the drawings shall be cause for rejection.

4.6.1.2 Tests. Each set shall be subjected to the applicable tests marked "X" under column 3 of table II. For generator only tests, the tests shall be conducted prior to assembly into the set. Failure of any test shall be cause for rejection.

4.7 Inspection schedule. The test schedules shall be as shown herein. Nonconformance to the applicable requirement paragraph in section 3 shall constitute failure of the test and rejection of the generator, or the set, as applicable. The Government reserves the right to reject the equipment for not

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meeting any requirement herein, even though not performing a test directly related to the specific requirement. The requirements in section 3 shall apply to all tests performed as a part of another test. If maintenance shutdown, as permitted in table I, servicing, adjustments, and replacement schedule, would be due to occur because of hours accrued during previous testing after the beginning of any test, the shutdown shall be postponed until completion of the test or until completion of 100 hours of operation, whichever is the lesser. Table II shall be used for first article and quality conformance inspection as specified in 4.3 and 4.6.

4.8 Test procedures. Tests shall be conducted in accordance with MIL-STD-705 and as specified herein. During all tests, except as otherwise specified in table II, the set shall use fuel and lubricants as specified (see 3.8.1). Test instruments shall be of the laboratory type and shall have been calibrated within 30 days of the start of testing and at 6-month periods thereafter. Instruments used in calibration should have at least five times the accuracy of the instrument being calibrated. Calibrated reference instruments of lesser accuracy than standard and which are not used for any other purpose may be used for the required periodic check of test instruments. Instruments shall be calibrated at the frequencies at which they are going to be used. Direct-reading instruments shall have at least 0.5 percent accuracy and shall be connected to indicate in the most accurate portion of their range. On dc instruments, the readings shall not be made on the lower 15 percent of the scale. On ac instruments, the readings shall not be made on the lower one-third of the scale. When the test methods call for the use of voltage and frequency recording-type meters other than oscillographs, the Texas Instrument Company model PDRHXFHXVA-A16-XT or the Gould part number 2108-2202-005542 shall be used. Oscillograph galvanometer frequency response shall be flat from dc to not less than 3,000 cycles per second. When recording meters are specified for any part of a test, turn on the recording meters prior to starting the warm-up period of the set and record continuously for the duration of the test. The recording meter shall be operated at a minimum speed of 6 inches per hour during the portions of the test where steady state loading conditions exist and shall be operated at a minimum speed of 6 inches per minute at least 30 seconds before, during, and after a load change. When test method 608.1 of MIL-STD-705 is performed load application steps shall be 0-1/2, 1/4-3/4, and 1/2-4/4, with each load being applied and removed four times. When test method 608.2 is performed the no load to rated load change shall be made by changing from no load to 1/2 load allowing 30 seconds for stabilization and changing from 1/2 load to full load. Individual tests requiring a series of data points shall be repeated if an interruption occurs during the test (e.g., data points being taken for plotting a curve, etc.).

4.8.1 Reliability test. The reliability test shall be conducted in accordance with test method 695.1, reliability test, of MIL-STD-705. The test shall be two generators, each run for 975 hours simultaneously. The accept/reject criteria for reliability shall be as specified in Test Plan XIVc of MIL-STD-781. This allows for a total of 5 reliability failures from the two sets. The diesel fuel, during the reliability test, shall conform to VV-F-800, with sulfer content controlled to between 0.55 and 0.65 percent. Both sets shall run 100 hours on JP-5, and the remaining 875 hours on the VV-F-800 fuel specified above. Fuel consumption shall be in accordance with 670.1 of test

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method 695.1.3.3.1.c of MIL-STD-705 for 0/4, 1/4, 2/4, 3/4, and 4/4 of rated load for both diesel and JP-5 fuel. See 6.3.15 for failure of definition.

4.8.2 High oil temperature protective device test. The test will be conducted using a simulated oil temperature bench test by immersing the temperature sensing element in oil and heating the oil. The test shall determine that warning and shutdown temperatures are within manufacturer's recommended temperature limits, that controls function properly for warning, shutdown, and reset startup operations.

4.8.3 Tiedown. In lieu of the test, an analytical proof and tiedown diagram may be substituted to show that the equipment will conform to the requirements.

4.8.4 Parallel operation test for production sets. Each production set shall be tested in parallel with the first article set(s). This test shall determine proper operation of all parallel operation controls and devices including the reverse power safety devices. It shall also check for proper load division and kilovolt amperes division at no load, half load, and full load. Set instrumentation may be utilized.

4.8.5 Annunciator alarm system. Test annunciator alarm system for correct operation and indication of all malfunctions.

4.8.6 Remote control and operation test. The remote control and operation test shall be conducted with set and remote control station interconnected by use of the 1000 foot cable specified. By remote operation, determine compliance of: Engine start and stop; voltage and frequency adjustment; application and removal of load; provision to monitor set and bus frequency and voltages, real and reactive power annunciation system; and parallel operation.

4.8.7 Automatic control system test. The automatic control system test shall be conducted using two complete generator sets, one automatic control module interconnected by a 100 foot cable. By simulating a power failure, determine that performance is in accordance with 3.18.

4.8.8 Remote operation test. The remote operation test shall be conducted by removing the control cubicle 500 feet away from the set. The set shall be started, loaded, and stopped three times. All controls shall be operated to determine proper operation.

4.8.9 Automatic control system test (sets designated for specific installations). When specified (see 6.2) interconnect all designated sets, and one automatic control module, connect the sets for parallel operation and determine compliance with 3.18, as follows:

a. The sets shall be preheated for at least six hours, the speed droop set zero and not more than two percent voltage regulation. The load shall be 0.8 PF and sufficient to completely load the sets.
b. Adjust the 120 volts to the automatic start circuit to 109 volts and to 131 volts and determine compliance at each condition.
c. Reduce the 120 volts to the automatic start circuit to 107 volts and determine that all sets (1) start, (2) reach the ready to load condition, (3) act to disable the utility circuit breaker.

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d. Determine which one of the sets was selected as the master set and that the critical load feed circuits are closed, loading the set to at least 90 percent rated load.
e. Determine that the other sets are automatically paralleled and the load increased to at least 90 percent of the combined KW rating of the connected sets.
f. Operate the sets in parallel at between 90 and 100 percent of combined kW for 8 hours, reduce the load to 50 percent of the combined kW rating of the sets and operate for 8 hours, gradually increase (small steps) the load to between 90 and 100 percent of the combined kW rating of the sets, in not less than one hour or more than two hours, and operate at this condition until the sets have operated in parallel for 24 hours.
g. Reduce the load by 500 kW and disconnect one set from the system and stop the set. When the system consists of three or more sets, operate the remaining sets in parallel for one hour.
h. Repeat step g until there is only one set operating. Disconnect the remaining set from the load and stop the set.
i. Repeat steps c through h after a 24-hour shutdown except that the set that was the master set shall be delayed sufficiently to permit another set to become the master set.
j. Repeat steps c through i with the speed droop at 3 percent and the voltage regulation at no more than 2 percent.
k. Adjust the frequency to the automatic start circuit to 58.5 Hz and 61.5 Hz and determine compliance at each condition.
l. Repeat steps c through j by reducing the frequency to the automatic start circuit to 57.5 Hz. Determine that the sets do not start until the preselected time delay has elapsed.

4.8.10 Voltage and frequency drift test. The voltage and frequency drift test shall be conducted immediately following the -25° F test. Operate the set at rated load at -25° F until the set is stabilized. Within 8 hours increase the ambient temperature 60F°. The increase in temperature shall not be less than 5F° (2.8C°) per hour. Continue operation of the set at rated load until the set has stabilized. No adjustments to the set shall be made for the duration of the test.

4.8.11 High temperature test for optional equipment.

4.8.11.1 Remote control. The remote control and remote control cable shall be used to start, load, and stop the set after the control and cable have been at 125° F for at least 24 hours. The start, load, and stop sequence shall be repeated 10 times. Load on the set shall be rated load. Load shall be maintained for at least 15 minutes for each of the first nine cycles; for the last cycle, the load shall be maintained for 2 hours.

4.8.11.2 Automatic control system. The automatic control system shall be used to start and parallel two sets after the controls have been in the 125° F ambient for at least 24 hours. The test shall consist of an automatic start, automatic assumption of rated load, and stop. This sequence shall be repeated 10 times.

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4.8.12 Low temperature test of optional equipment.

4.8.12.1 Remote control. Conduct the test described in 4.8.11.1 at -25° F (-31.7° C).

4.8.12.2 Automatic control system. Conduct the test described in 4.8.11.2 at -25° F (-31.7° C).

4.8.13 Overload test. Operate the set at rated load and frequency until the set is stabilized. Increase the load in one step to 110 percent of rated 60 Hz load and operate for two hours. Reduce load to rated load and operate for 30 minutes.

4.8.14 Humidity test of optional equipment. The remote control and automatic control system shall be subjected to the five 48-hour humidity cycles specified in 711.1. Within an hour after removal from the test chamber the remote control and automatic control system shall be used to start, load and stop a generator set.

4.8.15 Overspeed protection devices. Prior to installation on the engine the speed switch will be tested and calibrated. The calibration fixture will include a variable speed motor, speed measuring device, mounting device for the speed switch, cabling to activate indicating lamps. The contacts in the speed switch will be calibrated to function at 300 ± 45 RPM (SS1) 600 ± 45 RPM (SS2) ±, 1080 ± 45 RPM (SS3 overspeed switch). After the overspeed switch functions, the manual reset will be tested. The speed switch shall be installed on the generator set utilizing an extension harness incorporating a push button switch attached to the speed switch and the engine harness. The push button will be wired across the overspeed switch normally-open contacts. After speed switch installation and with the above extension harness properly connected to speed switch and set harness and with the generator set operating at rated speed and no load, depress the push button switch momentarily. This action should immediately cause engine shutdown. After shutdown, reset both air valves on the engine.

4.8.16 Altitude operation test (mountain method). If it is more convenient to transport the generator sets to the required altitude on a mountain, the altitude test can be performed at the actual required altitude in lieu of using a pressure chamber. The apparatus, procedure and procurement document requirements for the method are the same as for test method 720.1c of MIL-STD-705.

4.8.17 Air transportability test. One generator set shall be transported to a site specified by the USAF. The air transportability loading will be performed by USAF personnel. It is necessary to perform separate loading on each type of specified aircraft.

4.9 Inspection of packaging.

4.9.1 Quality conformance inspection. Each generator set shall be inspected for compliance with the requirements of MIL-G-28554 and against the additional defects listed in 4.9.2.

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4.9.2 Examination. Each packaged generator set shall be examined for the following major defects in addition to those applicable ones from MIL-G-28554.

163. Anti-freeze of other than specified type used for preservation of cooling system for levels A and C (5.1.1 and 5.1.2).
164. Engine preservation for level C not in accordance with the alternate procedures specified (5.2.2).
165. Generator sets without housings not protected with wood cover as specified for level A (5.2.1).
166. Openings of generator set housings, if so equipped not sealed with tape for level A as specified (5.2.1).
167. Generator sets not protected with a shroud for level C as specified (5.2.2).
168. Marking incorrect, illegible, or missing (5.3).

4.9.3 Rejected units. The generator set shall be rejected if the packaging does not comply with one or more of the major packaging defects. The set may be delivered if the deficient packaging is corrected to comply with specified requirements.

4.10 Comparison testing. The Government may select sets at any time during the contract production period and subject these sets to any examination and test specified herein, necessary to determine that the selected sets meet all requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Sets will be selected at random from those which have been accepted by the Government and will not include the previously inspected first article sets. Acceptance of a comparison set shall not exclude the remaining sets from the quality conformance inspection and acceptance provisons specified in section 4. In addition to any test specified as part of the comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the comparison and failure of such additional test shall have the same effect as failure of those tests specified as comparison.

4.10.1 Comparison failure. Failure of a comparison set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.10 shall be cause for rejection of the comparison set and shall be cause for refusal by the Government to continue acceptance of production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sets previously accepted and produced under the contract. Any deficiencies found as a result of the comparison will be considered prima facie evidence that all sets accepted prior to the completion of the comparison evaluation are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

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5.1.1 Level A. The complete generator set shall be preserved in accordance with the level A requirements specified on drawing 76-11458, sheet 7.

5.1.2 Level C. The complete generator set shall be preserved in accordance with the level C requirements specified on drawing 76-11458, sheet 7.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. The generator set, preserved as specified in 5.1, shall be enclosed with a cover as shown on drawing 76-11458, sheets 1-6. If the generator set is equipped with a housing, a wood cover is not required; all openings in the housing shall be sealed with waterproof tape conforming to MIL-T-22085, type II. All disassembled components and accessories shall be secured to the base of the generator set inside the cover or housing.

5.2.2 Level C. The preserved generator set shall be enclosed with a water-proof shroud secured to prevent water leaks and loosening due to wind. Suggested shroud material would be black polyethylene of at least 8 mils thick or a reinforced plastic film having sufficient strength for the application.

5.3 Marking. Marking of all interior and exterior packs shall be in accordance with MIL-STD-129.

5.4 Connection and torg. The set shall be connected for 120/208 volts, and the neutral output terminal (Lo) to ground stud conductor shall be connected when the generator set is shipped. A connection tag shall be attached to one of the lead terminals stating: "This set is connected for 120/208 volts.

6. NOTES

6.1 Intended use. The sets are intended to supply power for multipurpose use in military applications.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of the first article set to be furnished (see 3.2).
c. When the lead acid storage batteries other than as specified are required.
d. When batteries are to be shipped other than as specified.
e. When a housing kit is required (see 3.7).
f. When an automatic control module is required (see 3.18).
g. When a remote control module is required (see 3.19).
h. When a remote control cable is required (see 3.20).
i. When the length of the remote control cable shall be other than as specified (see 3.20).
j. When a special system test for sets designated for specific installations is required and number of sets to be tested as a system (see 4.8.9).
k. Color of finish coat required.
l. Levels of preservation and of packing required (see 5.1 and 5.2).

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m. When the Government will conduct any or all of the first article examination and tests, when the Government will conduct some but not all of the first article examinations and tests, the contracting officer should specify which examinations and tests will be conducted by the Government and which examinations and tests shall be conducted by the contractor (see 3.2).

6.3 Definitions.

6.3.1 Start. A set is considered to have started when it is operating at rated voltage and speed without the further use of starting aids.

6.3.1.1 Cranking cycle. Each cranking cycle shall consist of 15 seconds of continuous cranking, followed by a 30-second rest period, on an inactive set (engine fuel shutoff solenoid de-energized).

6.3.2 Standby operation. A condition during which a nonoperating set is kept warm and ready to start through use of integral electric winterization kits.

6.3.3 Temperature and humidity damage. Temperature and humidity damage is defined as corrosion, breakage, deformation, reduction of insulation resistance below 50,000 ohms or conditions causing malfunction of any component or part.

6.3.4 Dangerous flexural vibration. Dangerous flexural vibration is defined as a vibration which occurs at a speed at which maximum stress in the shaft from flexural vibration exceeds 9,000 psi.

6.3.5 Dangerous torsional critical speed. Dangerous torsional critical speed is defined as the speed at which maximum vibrating stress in the shaft from torsional vibration exceeds 5,000 psi.

6.3.6 Rough handling damage. Rough handling damage is defined as deformation, loosening, breakage, change of fit of any component or part, or any condition resulting in malfunctioning of the set.

6.3.7 Transportation. Transportation is defined as the moving of the set from one location to another by air, rail, highway, or water, including the loading and unloading operations required for the relocation.

6.3.7.1 Rail transportation. Rail transportation shall include shocks and stresses from impact at speeds up to and including 8 mph under test conditions specified in MIL-STD-705, method 740.5.

6.3.7.2 Highway transportation. Highway transportation, utilizing commercial, multiaxle trailers, shall include speeds up to and including 40 mph over paved highways, and at speeds up to and including 10 mph over unpaved roads.

6.3.7.3 Water transportation. Water transportation shall include transport by ship or barge, operating over any type of waterway.

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6.3.7.4 Air transportation. Air transportation shall include transport by C-130, C-141 or C-5A aircraft.

6.3.8 Weight and dimensions. The weights of the sets are dry weights and do not include the weights of fuel, coolant, lubricating oil, electrolyte, hydraulic oil and optional equipment. The dimensions shown do not include the optional equipment except for the housing kit.

6.3.9 Dripproof enclosure. A dripproof enclosure is an enclosure so constructed that falling drops of liquid or solid particles striking the enclosure at angles from 0 degrees to 15 degrees from the vertical cannot enter the enclosure either directly or by striking and running along a horizontal or inwardly inclined surface.

6.3.10 Frequency and voltage regulation. Frequency and voltage regulation is defined as the difference between the no load value and the rated load value divided by the rated load value.

6.3.11 Stable set operating conditions voltage and frequency. Stable set operating conditions are defined as the conditions specified for steady state performance.

6.3.12 Overshoot and undershoot. Reference MIL-STD-705, figure 608.1-VII for definitions of overshoot and undershoot.

6.3.13 Meter accuracy. Accuracy is a number which defines the limit of error expressed as percentage of full scale value. Error is the difference between the indication and the true value of the quantity measured. It is the quantity which, when algebraically subtracted from the indication, gives the true value. A positive error denotes that the indication of the meter is greater than the true value.

6.3.14 Fail-safe. A circuit arrangement so that an opening anywhere such as a broken wire or the operation of a circuit protection device, will interrupt the circuit output; or if a mechanical malfunction, will return the device to a condition to restrict the damage to the failed part(s).

6.3.15 Failure. A failure is defined as the inability of an item to perform within specified limits. The contracting officer will identify all failures and will classify them into the following categories.

6.3.15.1 Nonrelevant failure. A nonrelevant failure is one which is caused by any of the following:

1. Failure of test instrumentation or monitoring equipment that is external to equipment under test.
2. Damage resulting from accident or mishandling.
3. Failures due to procedural errors by the technicians or operators.
4. Secondary failures as defined by MIL-STD-781.
5. A failure (or shortcoming) that does not affect the intended purpose of the set (furnish electrical power within specified limits, or control the set as intended herein).

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6. Failures resulting from operating items beyond requirements, for example, if belts are run to failure to determine mean lift, failures after 1000 hours are not considered set failures.
7. Failures which may be corrected by normal operator functions, such as readjustment of voltage.

6.3.15.2 Relevant failure. All failures shall be considered relevant unless they are classified to be nonrelevant.

6.3.16 Altitude. Altitude is the elevation in feet above mean sea level (msl); the equivalent barometric pressure in millimeters of mercury (mmHg), shown in parenthesis following the altitude (i.e., 8000 feet [564.4 mmHg]), is based on a barometric pressure of 29.92 inches (760 mm) of mercury under assumed standard msl conditions of 59° F (15° C) and air mass density of 0.001225 grams per cubic centimeter.

6.3.17 Rainproof. An enclosure is considered rainproof when it is so constructed as to prevent any significant accumulation of water within the enclosure and no water has entered the enclosure at a level higher than the lowest live part.

6.3.18 Damage. Damage is defined as any failure (see 6.3.15), rough handling damage (see 6.3.6) or degradation in life. Fusing of a replaceable fuse is not considered damage, providing it is performing its intended function.

6.4 First article model. Any changes or deviations of production sets from the approved first article model during production will be subject to the approval of the contracting officer. Approval of the first article model will not relieve the contractor of his obligation to furnish sets conforming to this specification.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists and supplier's maintenance and operation manual to be furnished with each engine-generator set.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary including any special tools, and instructions on shipment of engine-generator sets.

6.7 Configuration management. The generator sets controlled by this specification are under the Configuration Management authority of the DOD Project Manager for Mobile Electric Power. All Engineering Change Proposals (ECPs), Requests for Waivers (RFWs), and Requests for Deviations (RFDs) must be processed in accordance with DLAR 4120.7, AR 700-101, AFR 400-50, NAVMATINST 4120.100A, MC011310.8C, Joint Operating Procedures, Management and Standardization of Mobile Electric Power Generating Sources.

6.8 Drawings. The drawings forming a part of this specification may contain more specific or stringent requirements not fully delineated in the test of the

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specification. This, however, is not a conflict and the drawing details govern (e.g., should the specification require a minimum of 750 hp and the drawings require a minimum of 835 hp; acceptance criterion is 835 hp). Where the duplication involves tolerances, the foregoing applies. In addition, if the dual entries are identical (e.g. ± 10 percent) this is to be interpreted as a reiteration and neither a cumulative nor multiplicative "tolerance on a tolerance: (e.g., acceptance criterion is ± 10 percent and neither ± 20 percent nor ± 11 percent).

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Air Force - 99

Review activities:
DLA - GS
AF - 80

Preparing activity:
Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-52880A	Generator Set, Diesel Engine, 500 KW 50/60 Hertz, Tactical

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
Utility	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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